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Research Progress and Trends of Deep Learning in Stock Price Prediction: A Systematic Review from LSTM to Transformer

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Abstract. Stock price prediction is a key issue in quantitative finance. Due to the characteristics of the financial market, traditional prediction methods often fail to achieve accurate and stable prediction results. In recent years, deep learning technology, especially architectures such as Long Short-Term Memory (LSTM) and Transformer, have stood out with their excellent and stable performance. This paper systematically reviews the update and iteration process of stock prediction methods and elaborates on the classification of mainstream deep learning models, analysing the characteristics, advantages and limitations of different models, and comparing the differences in data sets, indicators and performance in empirical studies. In addition, it further discusses challenges such as data noise, overfitting, interpretability and computational efficiency, and thereby looks forward to future research directions such as multimodal information fusion, interpretable artificial intelligence and real-time adaptive learning. This paper aims to provide a complete technical roadmap for researchers in the field of stock price prediction to apply deep learning methods systematically.

1 Introduction

The accuracy of stock price prediction is one of the ultimate goals in modern finance, directly determining the success or failure of quantitative investment strategies and the effective management of downside risks. Stock prices are influenced by various factors such as the macroeconomic environment, industry development prospects, company operating conditions, and market participants' sentiments, exhibiting significant characteristics such as high volatility, non-linearity, and non-stationarity, which greatly increase the difficulty of predicting stock prices. Therefore, how to construct a model that combines accuracy and stability to predict the complex and variable stock market has become the long-term research goal for relevant personnel.

Early stock predictions mainly utilized models such as ARIMA and GARCH, which are based on linear assumptions and applicable to stationary sequences. However, these models have limitations such as difficulty in capturing nonlinear features and sensitivity to noise [1]. Subsequently, machine learning techniques such as Support Vector Machines (SVM),

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Random Forests (RF), and eXtreme Gradient Boosting (XGBoost) were widely applied in stock market predictions. These methods improved prediction accuracy and had stronger capabilities in capturing nonlinear relationships and extracting data features, but they relied on human assumptions and economic theories [2].

At the deep learning stage, based on the Recurrent Neural Network (RNN), optimization models such as LSTM and Gated Recurrent Unit (GRU) effectively overcome the problems of gradient disappearance and explosion through the gating mechanism, and can capture and predict the corresponding short-term dynamic features [1]. The emergence of LSTM is of epoch-making significance, but when dealing with sufficiently long sequences, it still inevitably experiences information attenuation, and its sequential processing mechanism also leads to low computational efficiency and other problems. Another issue is that LSTM models are modeled along the time axis within the same sequence, resulting in relatively weak modeling of cross-sequence relationships. This pain point stimulated and drove the development and transformation of technology. The Transformer model, due to its success in the sub-field of artificial intelligence and its excellent self-attention mechanism, was eventually introduced into stock prediction. Its emergence successfully solved the bottleneck problem of LSTM and brought a new solution to stock prediction. Moreover, researchers are no longer satisfied with using a single model for prediction and are committed to complementing the advantages of each model to build a more powerful hybrid model, such as LSTM-Transformer, to seek another breakthrough in model performance and prediction accuracy.

Although some studies have achieved remarkable results and there are several related review papers that have explored the application of deep learning in the financial field, there is still a lack of a systematic review that specifically focuses on predicting stock prices, systematically summarizes the development process from LSTM to Transformer hybrid architectures, and also takes into account the analysis of model principles and empirical comparisons. Therefore, this paper aims to fill this gap: systematically review the evolution process of stock prediction models, in detail analyze the specific characteristics and advantages and disadvantages of mainstream deep learning models, summarize the existing research results and compare the similarities and differences among models, discuss the current problems and future development directions.

2 Overview of Mainstream Technologies

The trend of deep learning models gradually becoming the mainstream in the field of stock prediction has been observed. The following content introduces several mainstream models: recurrent models, convolutional models, Transformer-related models, hybrid models, and some extension methods. At the same time, the corresponding characteristics, advantages, and disadvantages are elaborated.

2.1 RNN and Its Variants

RNN use a hidden state recurrence mechanism to remember historical information and handle the temporal correlation of sequential data. Their improved model, LSTM, introduces forget gates, input gates, and output gates, effectively solving the gradient problem of long sequence correlations that traditional RNNs have, and has significant advantages in capturing medium and short-term dependencies. Moreover, Fischer & Krauss' research shows that the performance of LSTM is significantly better than the memoryless model, so LSTM is widely used in financial market predictions [1].

Additionally, GRU achieves similar performance to LSTM with a simpler gating structure and is also widely used. Rahmadeyan's research shows that GRU is significantly more efficient in terms of computational efficiency than LSTM, but for different time periods, the prediction accuracy of GRU is slightly better or worse than LSTM [3]. However, LSTM/GRU is limited by the recursive calculation mode and the limited memory window. When facing issues such as ultra-long time spans, multi-sequence modelling, and market state mutations, their performance is still relatively weak.

2.2 Convolutional Neural Network (CNN) and CNN-LSTM Hybrid Model

CNN initially performed well in computer vision. Its local receptive field and hierarchical feature extraction mechanism were later applied to time series prediction tasks, used to extract local trends and cycles, etc., of stock market information, while filtering out interfering noise. Among them, when the time series is regarded as a one-dimensional image (1D image), CNN extracts multi-scale filtering features in a data-driven manner [4]. Then, Mehtab and Sen proposed to use CNN as the front-end encoder to read sequence data and extract key features, and LSTM as the back-end decoder to decode and predict the sequence, complementing the local perception of CNN and the temporal dependency of LSTM, forming a CNN-LSTM hybrid architecture for predicting the NIFTY 50 index value [5]. The results proved that this hybrid structure performed better in multi-step prediction. However, the hybrid model inevitably brings problems such as complex structure, many hyperparameters, and has defects such as overfitting and being sensitive to abnormal fluctuations and sudden changes.

2.3 Transformer and Its Variants

The Transformer architecture was initially applied in Natural Language Processing (NLP), and it does not rely on traditional recursion and convolution. It achieves global information interaction through a unique self-attention mechanism (Self-Attention). Subsequently, due to its strong parallel computing and long-range dependency modelling capabilities, Transformer was introduced into time series prediction to leverage its advantages, for capturing long-range dependency features and modelling cross-asset correlations. In recent years, many stock prediction models based on Transformer have emerged. For example, the Stockformer architecture for multi-variable time series prediction tasks converts the input variables into a single-variable output [6]. There are also studies evaluating the performance of different structured Transformers (only encoder, only decoder, encoder + decoder (standard), standard without embedding layer, standard with ProbSparse Attention), and the experimental results show that the Transformer model with only the decoder structure performs the best [7]. However, at the same time, Transformer also has the problem of financial sample quantity being small, which is prone to overfitting. Its self-attention mechanism inevitably brings issues such as high computational cost and high existing occupancy, amplifying the impact of errors and noise in abnormal situations.

2.4 Hybrid Models and Other Extension Methods

To mitigate the shortcomings of a single model, the extension of model structure and the mixture of models have been widely discussed.

One type is the direct fusion of LSTM and Transformer, which includes mechanisms such as cascading, parallel connection, and plug-and-play attention. Here, LSTM is responsible for local short-term information, while Transformer is responsible for global long-term information. For example, Zhang and Li adopted a dual-branch structure of LSTM-Transformer hybrid model to achieve multimodal fusion prediction of price and sentiment information, and the prediction accuracy was significantly higher than that of a single model

[8]. Another type is based on Transformer, introducing frequency domain decomposition and time encoding mechanisms to simultaneously capture the periodic and non-periodic components in the time series. For instance, Zhang and Li proposed the CEEMD-Time2Vec-Transformer (ETT) stock prediction model, which significantly enhanced the prediction performance compared to other benchmark deep learning models [9]. Additionally, graph neural networks (GNN) have also been widely discussed. Zhang et al. mentioned the Chart GCN, which converts sequence important nodes into graphs and uses GNN to compare their similarities [10]. There is also the Attribute-Driven Graph Attention Network (AD-GAT), which makes predictions through momentum overflow. Stock prediction models are being optimized with the aims of cross-model fusion, multimodal fusion, and interpretability to significantly improve the model's usability.

3 Applications and Empirical Research

3.1 Dataset and Sample Description

At the data source level, commonly used datasets include national or regional stock market indices (such as the S&P 500 in the United States, NIFTY 50 in India, and the Moroccan Stock Exchange Index), industry sector indices (such as technology, energy, and finance), and multi-asset portfolio or individual stock sample data. In terms of time scale, daily data (daily data) is often used for medium and long-term trend prediction due to its lower noise and stable trend. Lahebb and Benaly compared the use of ARIMA, LSTM, and Transformer models to analyse the daily data of three well-known credit companies in Morocco and made short-term investment decisions [11]. In contrast, minute-level or high-frequency data (high-frequency data) can reveal market microstructure and is often used to analyse short-term market dynamics, imbalance in order flow, and price shock mechanisms. Zhang et al. used Limit Order Book (LOB) data and developed a convolutional filter-LSTM model to explore the price dynamics under the microstructure [12].

In recent years, data structures have been continuously improved and are gradually evolving towards multi-dimensional and multi-modal directions. At the same time, in addition to traditional sequences such as prices and trading volumes, researchers have begun to introduce exogenous features such as sentiment variables, macroeconomic and technical indicators to enhance the interpretability and generalization ability of the models.

3.2 Relevant Evaluation Criteria

In the review by Saberironaghi et al., four categories including overall, classification, regression, and profitability were mentioned, along with a total of 12 evaluation indicators [13]. This comprehensive and in-depth assessment of the deep learning model ensures its usability and the accuracy of the prediction results. Depending on different research goals, researchers typically use the following indicators to evaluate the relevant performance of the deep learning model:

• Root Mean Squared Error (RMSE): Reflects the overall deviation between the model's predicted values and the actual values. The smaller the value, the better the fitting effect, and it is more sensitive to large deviations.

• Mean Absolute Error (MAE): Is the average deviation between the model's predicted values and the actual values, reflecting the stability of the model's prediction.

•Mean Absolute Percentage Error (MAPE): Is the average percentage error between the model's predicted values and the actual values.

• Directional Accuracy (DA): Reflects the accuracy of the model in predicting the direction of stock price changes, directly corresponding to the actual usability of the model in investment decisions.

• R^2/ Adjusted R^2: Indicates the model's ability to explain the variation of independent variables, also reflecting the fitting degree of the model, but only as a descriptive indicator.

• Sharpe Ratio: Reflects the relationship between returns and risks. The higher the value, the higher the return under the given risk.

3.3 Empirical Results Comparison and Trends

Based on multiple empirical studies, deep learning models outperform traditional machine learning and statistical methods in stock price prediction.

The LSTM model and its variants have been extensively proven to be reliable benchmark models, especially for medium and short-term prediction tasks. In a study by Fischer and Krauss on the prediction of all S&P 500 index constituents from 1992 to 2015, the daily return rate of LSTM was 0.46%, significantly higher than the standard DNN at 0.32% and logistic regression at 0.26%, among other non-memory classification methods [1]. Moreover, LSTM has a significant advantage in generating trading signals with direct economic value, with a Sharpe ratio of up to 2.34, while most other control methods are far less than 1.0 [1]. Subsequently, the Transformer model was further optimized based on LSTM, especially in long sequences, demonstrating the superiority of the self-attention mechanism in long-range dependencies and cross-asset correlations. Wang et al. predicted the stock prices of new energy vehicle enterprises and mentioned that the Transformer model has better prediction accuracy compared to CNN, RNN, and LSTM traditional deep learning models, with an average MAE decrease of approximately 20.73%, MSE decrease of approximately 34.84%, and MAPE decrease of approximately 25.63%, all demonstrating smaller errors and better fit [14]. However, in small sample scenarios, LSTM has stronger stability due to its simple model architecture. Nevertheless, the hybrid model LSTM-Transformer has become the focus of research. Many research results show that by integrating the local dynamic capture of LSTM and the global dependency modelling of Transformer, it achieves comprehensive optimization based on the parent model. Zhao et al. mentioned that the MAE, RMSE, etc. indicators of the LSTM-Transformer hybrid model are reduced by more than 50% compared to LSTM and Transformer models, indicating lower errors and more stable models [15]. At the same time, the R^2 value of the hybrid architecture (0.9618) is higher than that of the parent model (LSTM: 0.8430; Transformer: 0.7763), representing stronger explanatory power [15].

From the overall trend, LSTM has a significant advantage in direction determination and generating trading signals with high Sharpe ratios, while the Transformer model further improves prediction accuracy; the LSTM-Transformer hybrid architecture simultaneously has stronger interpretability based on higher accuracy and stability. In summary, stock prediction models have evolved in recent years from LSTM → CNN-LSTM → Transformer → hybrid model → multimodal fusion model, with increasing complexity of architectures and fusion dimensions, reflecting the design paradigm from single model optimization to collaborative integration.

4 Discussion and Outlook

Although deep learning models have made significant progress and breakthroughs in stock price prediction in recent years, there are still many challenges and difficulties between theoretical knowledge research and the application of actual models. The following will analyze the current problems from three levels: data, model, and deployment application, and propose future development trends and research directions.

4.1 Current Challenges

4.1.1 Data Level

Firstly, financial market data has extremely low signal-to-noise ratio and jump characteristics, which causes the model to easily overfit historical noise rather than learning the patterns of stock price fluctuations; Secondly, changes in macroeconomic policies and black swan events can lead to a significant decline in the generalization ability of models based on historical data in new environments; At the same time, problems such as the lag of up and down labels and the unidirectional data in volatile markets pose challenges to supervised learning; Moreover, continuous bull or bear markets can lead to unequal sample sizes, causing the prediction to be biased towards the majority category, and the data learned by the model usually does not include vanished companies, and the beautified data can also affect the prediction accuracy; Finally, how to deeply integrate diverse information such as news, social media, financial reports, and macroeconomics is the key to breaking through the performance bottleneck of existing models.

4.1.2 Model Level

Firstly, the "black box" nature of deep models makes it impossible to provide the corresponding basis for making judgments, although post-event explanation techniques (attention weight visualization, SHAP values, LIME) are attempting to solve this problem, there is still a lack of a unified solution; Secondly, the available samples for financial prediction (N) are very few, but the models have a large number of parameters (P), this "N << P" situation allows the model to have sufficient space to memorize the noise and specific patterns in the training data; Moreover, the self-attention mechanism of the Transformer architecture associates elements in the sequence pairwise, and the huge computational load and memory usage in long sequences can cause some delays, affecting efficiency; Additionally, the design of hybrid models lacks theoretical basis guidance, and their performance improvement is mostly based on researchers' intuition and continuous trial and error, which is extremely costly and difficult to reproduce; Finally, existing models mostly follow the principle of offline training and static prediction, which is contrary to the dynamic changes of the financial market, resulting in the gradual decline of model performance over time.

4.1.3 Deployment and Application Level

Firstly, in scenarios such as high-frequency trading and decision implementation, there are extremely strict requirements for model running speed, and deep models often cannot meet these requirements; Secondly, if a financial crisis, epidemic, or other emergencies occur, the general patterns learned by the model instantly become invalid, and the prediction performance drops significantly; Moreover, a single prediction model cannot directly create profits, and a complete closed loop of prediction-decision-execution-risk control is needed to exert its value; Finally, the "black box" characteristics of deep models and the transparency, explainability, fairness, bias control, and stability requirements of regulatory authorities are contradictory, which is not conducive to their development.

4.2 Future Trends and Research Directions

Looking forward, first, it is not limited to simple feature concatenation. By integrating diverse information on the Transformer architecture to build a large model in the financial field that can associate cross-modal information; Then, integrate the causal inference framework with deep learning to enable the model to complete the transformation from "black box" prediction to "glass box" diagnosis, to strengthen the credibility of the model's prediction results; At the same time, add AutoML and NAS technologies to enable the model to achieve automated search and automatically adjust the model architecture according to the current market regime, to adapt to increasingly complex hybrid models and massive hyperparameters; Furthermore, the design enables real-time learning from the data stream without the need for re-training, and can quickly adapt to new environments, optimizing parameters rapidly using only a small number of samples, thus enabling the model to have the ability for continuous learning; on this basis, introducing RL agents for decision-making, and the deep model to play the role of perceiving the environment, forming a complete closed loop from market perception to investment decision-making, achieving the transformation from assisting human decision-making to autonomous agent-driven; further, introducing Linear Transformer, Performer and other linear attention mechanisms, while exploring S4, Mamba and other state space models, fundamentally changing the computational paradigm of Transformer to solve the bottleneck of low efficiency in processing long sequences; finally, using transfer learning and meta-learning techniques, transferring the knowledge and patterns learned by the model in a certain market environment to a completely new market with less historical data, to quickly obtain a highly performing predictive model.

5 Conclusion

With the continuous development of the field of deep learning, research on stock price prediction based on LSTM, Transformer and their hybrid models has achieved significant results. This paper systematically reviews the model development process from RNN to hybrid Transformer architectures, analyzes in detail their principles, advantages and disadvantages, and application scenarios, and collates many empirical studies, comparing their performance on different datasets and evaluation metrics. Finally, this paper also discusses the current problems faced by the technology and future research directions.

Overall, LSTM has advantages in short-term prediction, while Transformer dominates in long-term dependence and cross-asset modeling. Hybrid models and multimodal fusion have become an inevitable development trend. Future research should focus on breakthroughs in interpretability, real-time performance and adaptive learning, to achieve a perfect balance between accuracy, efficiency and robustness.

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